Compound interest tables

1 $\$3900$ is invested for three years at a rate of 10% p.a., compounded annually.

a Complete the table below to determine the final value of the investment:

	Balance at beginning of year	Interest earned
First year	$\$3900$	$\$390$
Second year	$\$4290$	$\$429$
Third year		
Fourth year		–

b Calculate the total interest earned over the three years.

2 $\$3700$ is invested for three years at a rate of 7% p.a., compounded annually.

a Complete the table below to determine the final value of the investment:

	Balance at beginning of year	Interest earned
First year	$\$3700$	$\$259$
Second year	$\$3959$	$\$277.13$
Third year		
Fourth year		–

b Calculate the total interest earned over the three years.

3 $\$3000$ is invested at 4% p.a., compounded annually. The table below tracks the growth of the principal over three years:

Time Period (years)	Value at beginning of time period	Value at end of time period	Interest earned in time period
1	$\$3000$	A	B
2	C	$\$3244.80$	D
3	$\$3244.80$	$\$3374.59$	E

a Find the value of:

i A ii B iii C iv D

v E

4 $\$9000$ is invested for three years at a rate of 5% p.a. compounded annually.

a Complete the table:

	Interest (\$)	Balance (\$)
Starting balance	0	9000
After one year		
After two years		
After three years		

b Calculate the total interest accumulated over three years.

c Calculate the value of the investment at the end of the three years.

5 Maria invested $\$1400$ at 10% p.a., compounded annually over 3 years. Without using the compound interest formula calculate:

a The interest earned for the first year.

b The balance after the first year.

c The interest earned for the second year.

d The balance after the second year.

e The interest earned for the third year.

f The balance after the third year.

g The total amount of interest earned over the three years.

h The interest as a percentage of the initial investment, correct to one decimal place.

i The interest earned after three years if the investment was simple interest rather than compound interest.

j Which type of interest is best for this investment and by how much is it better.

6 The following compound interest table shows the final value of a $\$1000$ investment, for various interest rates, compounded annually over various numbers of years:

	5 years	10 years	15 years	20 years	25 years
$r = 5\%$ p.a.	1628.89	1790.85	1967.15	2158.92	2367.36
$r = 6\%$ p.a.	2078.93	2396.56	2759.03	3172.17	3642.48
$r = 7\%$ p.a.	2653.30	3207.14	3869.68	4660.96	5604.41
$r = 8\%$ p.a.	3386.35	4291.87	5427.43	6848.48	8623.08
$r = 9\%$ p.a.	4321.94	5743.49	7612.26	10 062.66	13 267.68

If $\$50\,000$ is invested and earns interest at 6% p.a. over 15 years, calculate:

a The value of this investment.

b The amount of interest earned.

7 The following compound interest table sh[redacted]he final value of a $1 investment, for various interest rates, compounded annually over various numbers of years:

	10 years	11 years	12 years	13 years	14 years	15 years
$r=8\%$	2.1589	2.3316	2.5182	2.7196	2.9372	3.1722
$r=9\%$	2.3674	2.5804	2.8127	3.0658	3.3417	3.6425
$r=10\%$	2.5937	2.8531	3.1384	3.4523	3.7975	4.1772
$r=11\%$	2.8394	3.1518	3.4985	3.8833	4.3104	4.7846
$r=12\%$	3.1058	3.4785	3.896	4.3635	4.8871	5.4736

After how many years will a sum of money triple in value if it is invested at 10% p.a., compounded annually?

Compound interest formula

8 Find the future value of the following:

a An investment of $\$2000$ earns interest at 6% p.a, compounded annually over 4 years.

b An investment of $\$1000$ earns interest at 2% p.a., compounded semiannually over 3 years.

c An investment of $\$8030$ earns interest at 3% p.a., compounded annually over 20 years.

d An investment of $\$4490$ earns interest at 2.8% p.a., compounded semiannually over 6 years.

e An investment of $\$8030$ earns interest at 3% p.a., compounded quarterly over 12 years.

f An investment of $\$9450$ earns interest at 2.6% p.a., compounded monthly over 14 years.

g An investment of $\$1710$ earns interest at 2.2% p.a., compounded weekly over 6 years.

h An investment of $\$3000$ earns interest at 4.5% p.a., compounded daily over 5 years. Assume one leap year over this period.

i An investment of $\$392$ earns interest at 2% p.a., compounded annually for 7 years. After this time the principal plus interest is reinvested at 4% p.a., compounded annually for 5 more years.

9 Maria has $\$1000$ to invest for 4 years and would like to know which of three investment plans to choose:

- Plan 1: invest at 4.98% p.a. interest, compounded monthly.
- Plan 2: invest at 6.44% p.a. interest, compounded quarterly.
- Plan 3: invest at 5.70% p.a. interest, compounded annually.

a Calculate the future value of each investment plan.

b State which plan Maria should choose [illegible]aximum return on her investment.

10 Hannah put $\$600$ in a savings account with interest compounded quarterly at a rate of 1.1% p.a. Calculate the amount that is in Hannah's account after a period of 21 months.

11 Tina has $\$900$ in a savings account which earns compound interest at a rate of 2.4% p.a.

If interest is compounded monthly, how much interest does Tina earn in 17 months?

12 John borrows $\$6000$ from a loan shark at a rate of 20% p.a. compounded annually. If he is not able to make any repayments, calculate how much John will owe at the end of 5 years.

13 Emma borrows $\$7000$ from a loan shark at a rate of 4.7% p.a. compounded annually. If she is not able to make any repayments, calculate how much Emma will owe at the end of 3 years.

Calculate the principal

14 Scott wants to have $\$1500$ at the end of 5 years. If the bank offers 2.3% p.a. compounded annually, how much should he invest now?

15 Tom wants to put a deposit on a house in 4 years time. In order to finance the $\$12\,000$ deposit, he decides to put some money into a high interest savings account that pays 5% p.a. interest, compounded monthly. If P is the amount of money that he must put into his account now to accumulate enough for the deposit, find P.

16 Ursula has just won $\$30\,000$. She decides to invest some of her winnings into a retirement fund which earns 8% interest p.a., compounded yearly. When she retires in 29 years, she wants to have $\$52\,000$ in her fund.

How much of her winnings should Ursula invest now to achieve this?

17 Beth's investment into a 12-year 4.4% p.a. corporate bond grew to $\$13\,190$.

Calculate the size of Beth's initial investment if the interest was compounded:

a Annually
b Semiannually
c Quarterly
d Monthly
e Weekly
f Daily

18 Victoria has been promised an inheritance of $\$70\,000$ in 5 years time. What is the most she can borrow now at a rate of 7% p.a. compounded annually, and still be able to pay off the loan with her inheritance?